The energy that becomes thermal

antocciphysics | TEACHER KEY | CPA

Learning target: Account for mechanical energy changes through heating and deformation.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 15; 3 course days later.*
75 J; spring energy scales with compression squared.

02. Explain a worked example (worked)
Opposing arrows; mechanical energy decreases; box-floor thermal energy increases under stated assumptions.

03. Look and explain (model)
Yes; 20=14+6.

04. Build the solution (practice)
Work -15 J; thermal gain 15 J if no other energy changes.

05. Use the idea (reason)
No; mass, material/specific heat, phase and losses matter.

06. Choose a method (mixed) *Review: Unit 4, Day 6; 12 course days later.*
No. Work and kinetic-energy change must refer to the same endpoints. Measure matching intervals.

07. Exit check (exit)
2 J; must seek another transfer/storage or measurement error.

08. STRETCH: Apply it another way (stretch)
Distance 5 m; thermal increase 20 J if friction is the only energy transfer.

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01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 15; 3 course days later.*
$|x|=\sqrt{(2\ U/k)}=0.05$ m; $|F|=15$ N.

02. Explain a worked example (worked)
Opposing arrows; mechanical energy decreases; box-floor thermal energy increases under stated assumptions.

03. Represent and interpret (model)
Yes; $20=14+6$.

04. Solve and justify (practice)
$25-4=21$ J; internal increase 21 J.

05. Analyze the reasoning (reason)
No; mass, material/specific heat, phase and losses matter.

06. Choose a method (mixed) *Review: Unit 4, Day 6; 12 course days later.*
W=triangle area=2.4 J; $v_f\approx3.464$ m/s.

07. Exit check (exit)
9 J; 6 N.

08. STRETCH: Apply it another way (stretch)
Height 2 m; gravity work 60 J, friction -20 J; K 40 J; $v=\sqrt{(80/3)}\approx5.16$ m/s.